

Experiment - 16

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Course Code: MLA0305

Slot: B

Aim

To develop a reinforcement learning application using **TensorFlow/Keras** and evaluate its performance using episode rewards.

Algorithm / Procedure

1. Initialize the CartPole reinforcement learning environment.
2. Create a PPO-based reinforcement learning agent.
3. Train the agent for a fixed number of episodes.
4. Test the trained agent in the environment.
5. Record the reward obtained in each episode.
6. Calculate the average and maximum rewards.
7. Plot the performance of the trained agent.

Code

```
!pip -q install gymnasium stable-baselines3
```

```
import gymnasium as gym
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
from stable_baselines3 import PPO
```

```
env=gym.make("CartPole-v1")
```

```
model=PPO("MlpPolicy",env,verbose=0)
```

```
model.learn(total_timesteps=3000)
```

```
rewards=[]
```

```
for episode in range(10):
```

```
    state,_=env.reset()
```

```
    total_reward=0
```

```
    done=False
```

```
while not done:

    action,_=model.predict(state,deterministic=True)
    state,reward,terminated,truncated,_=env.step(action)

    done=terminated or truncated
    total_reward+=reward
    rewards.append(total_reward)

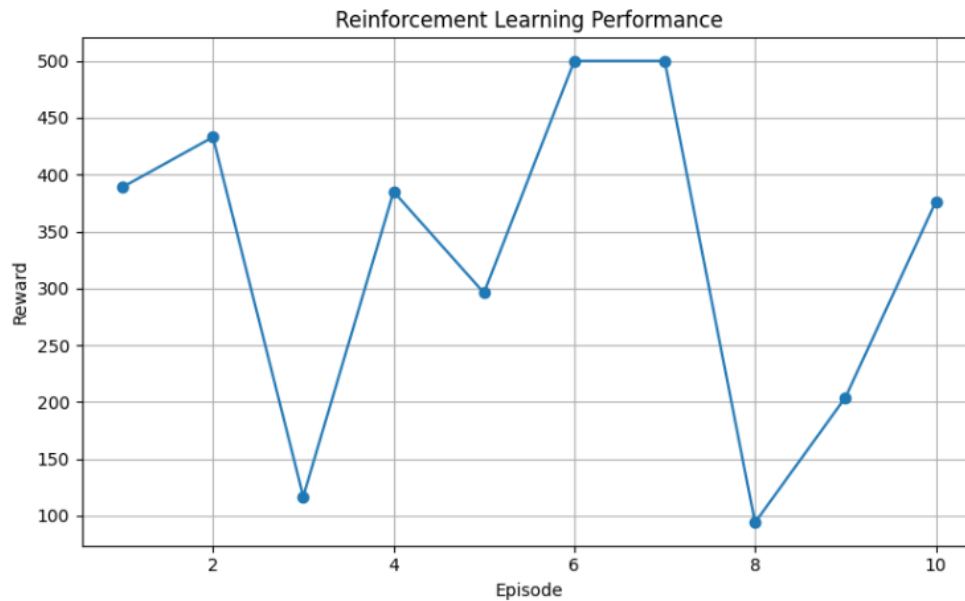
    print("Episode:",episode+1,"Reward:",total_reward)
print("\nAverage Reward:",round(np.mean(rewards),2))
print("Maximum Reward:",max(rewards))

plt.figure(figsize=(8,5))
plt.plot(range(1,11),rewards,marker="o")
plt.xlabel("Episode")
plt.ylabel("Reward")
plt.title("Reinforcement Learning Performance")
plt.grid(True)
plt.tight_layout()
plt.show()
env.close()
print("Experiment 16 Completed Successfully!")
```

Output

```
*** Episode: 1 Reward: 389.0
Episode: 2 Reward: 433.0
Episode: 3 Reward: 117.0
Episode: 4 Reward: 385.0
Episode: 5 Reward: 296.0
Episode: 6 Reward: 500.0
Episode: 7 Reward: 500.0
Episode: 8 Reward: 94.0
Episode: 9 Reward: 204.0
Episode: 10 Reward: 376.0
```

Average Reward: 329.4
Maximum Reward: 500.0



Experiment 16 Completed Successfully!

Result

The reinforcement learning application was successfully developed and evaluated. The reward graph demonstrates the performance of the trained agent across different episodes.